SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Transport Management System (TMS) – Mobile + Web

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1. Introduction

1.1 Purpose

This document specifies the requirements for the Transport Management System (TMS), a mission-based, card-driven platform for planning, assigning, executing, and auditing institutional transport operations involving faculty, guests, vehicles, and drivers.

1.2 Scope

TMS begins processing after a transport request card is assigned to the Transport Department from an external Task Management System. TMS manages request detailing, approvals, scheduling, vehicle/driver assignment, journey visualization, notifications, and audit logs.

1.3 Definitions

- **Card**: A transport request artifact received from the external system.
- **Mission**: A transport request that may include multiple journeys/legs/stops.
- **Leg/Journey**: A segment of travel (e.g., Institute → Airport).
- **Date Bucket**: Grouping of cards by travel date (Today/Tomorrow/Future).
- **Override**: Manual admin action that bypasses standard validations with audit logging.

1.4 Stakeholders

- Transport Department
- Faculty / Requesters
- Drivers
- Institute Administration
- IT / Engineering Team

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2. Overall Description

2.1 Product Perspective

TMS integrates with an external task system for intake and provides dedicated web and mobile interfaces for operations and notifications.

2.2 Design Philosophy (Non-Negotiable)

- Card-based workflow
- Mission-based transport (not single trip)
- Vehicles and Drivers treated as schedulable “users”
- Human decision override is always available
- Visual journey representation for operational clarity

2.3 User Classes and Characteristics

- **Faculty (Requester)**: Views/submits details; tracks status; receives notifications.
- **Transport In-Charge (Admin Operator)**: Reviews, approves, schedules, assigns, overrides.
- **Driver**: Has duty calendar; receives assignments; can apply for leave.
- **Vehicle (Resource User)**: Has capacity, type, availability calendar.

2.4 Operating Environment

- Web application (Transport Ops + Admin)
- Mobile application (Faculty + Driver; Admin optional)
- Real-time notifications (push/email/SMS configurable)
- Role-based access control

2.5 Constraints

- Must accept only approved/assigned cards from external system
- Must support multi-leg, multi-vehicle, multi-day missions
- Must allow manual override with full audit

2.6 Assumptions & Dependencies

- External Task Management System provides card assignment
- User identity and roles are available (SSO or user directory optional)

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3. System Features & Functional Requirements

3.1 Intake & Card Lifecycle

FR-3.1.1 TMS shall ingest transport cards assigned from the external system.
FR-3.1.2 TMS shall maintain card states:

- Assigned to Transport
- Details Pending
- Submitted to Transport Dept
- Transport Approved
- Scheduled (Date Bucketed)
- Vehicle Assigned
- Driver Assigned
- Active/In Progress
- Completed/Closed
- Rejected/Returned (optional)

3.2 Faculty Module

FR-3.2.1 Faculty shall view assigned transport cards and their statuses.
FR-3.2.2 Faculty shall fill/submit transport details:

- Travel date(s)
- Pickup/drop locations
- Passenger/guest details
- Passenger count
- Onward/return/one-way/multi-day flags
- Notes/special requirements

  FR-3.2.3 Faculty shall receive notifications for approval, assignment, and timing changes.

  FR-3.2.4 After Transport approval, Faculty view becomes read-only except for clarifications.

3.3 Transport Department – Incoming Requests

FR-3.3.1 Transport dashboard shall list new/pending/emergency cards.
FR-3.3.2 Transport In-Charge shall Approve, Reject, or Request Clarification.
FR-3.3.3 On approval, card shall be moved to date bucket based on travel date.

3.4 Date-Based Scheduling

FR-3.4.1 Cards shall be grouped under Today/Tomorrow/Future date buckets.
FR-3.4.2 Cards shall be sortable/filterable by urgency, location, passenger count, etc.

3.5 Vehicle–Driver Mapping (Daily Planning)

FR-3.5.1 System shall provide Vehicle Roster page showing:

- Vehicle ID/type/capacity/status
- Mapped driver
- Availability calendar
  FR-3.5.2 Drivers shall be able to submit leave requests.
  FR-3.5.3 Approved driver leave shall free the vehicle's driver slot.
  FR-3.5.4 Day closure shall normally require vehicle–driver mapping confirmation.
  FR-3.5.5 Admin Override shall allow day closure with idle vehicles; override must be audited.

3.6 Trip/Mission Assignment (Core)

FR-3.6.1 Admin shall assign vehicle(s) to a card via Assign Vehicle action.
FR-3.6.2 Vehicle selection shall display:

- Eligible vehicles by capacity/type
- Free vehicles
- Already assigned vehicles
- Per-vehicle timeline (start, destination, return times)
  FR-3.6.3 System shall allow reusing already assigned vehicles subject to admin decision.
  FR-3.6.4 System shall support timing adjustments and route merging.
  FR-3.6.5 System shall support multi-vehicle assignments across different legs.
  FR-3.6.6 On timing/assignment change, notifications shall be sent to stakeholders.

3.7 Journey Visualization (Mandatory UX)

FR-3.7.1 Each card shall show visual onward and return legs with icons and timestamps.
FR-3.7.2 Multi-stop routes shall be supported with ordered stop lists.

3.8 Supported Scenarios

System must support:

- Campus → Location → Campus
- Airport pickup/drop without faculty travel
- Multi-pickup/multi-drop
- One-way drop with independent return
- Overnight or multi-day vehicle stays outside campus
- Vehicle swap mid-mission
- Driver swap mid-mission
- Emergency re-scheduling

- Vehicle breakdown with replacement assignment

3.9 Notifications

FR-3.9.1 Faculty: approval, vehicle assignment, timing changes, driver info (optional)
FR-3.9.2 Driver: duty assignment, route details, time changes
FR-3.9.3 Admin: driver leave alerts, conflicts, emergency requests
Notification channels configurable (push/email/SMS).

3.10 Audit & History

FR-3.10.1 System shall store:

- Original request details
- All edits and time changes
- Vehicle/driver assignment history
- Override actions
- User/action timestamps

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4. Data & Domain Model (Conceptual)

4.1 Core Entities

- User (Faculty/Admin/Driver)
- Vehicle (type, capacity, status)
- Card/Mission (request + state)
- Journey/Leg (start, end, times, vehicle, driver)
- Stop (ordered location within a leg)
- Assignment (vehicle/driver to leg)
- Calendar (per user/vehicle/driver)
- Notification
- Leave Request (Driver)
- Audit Log

4.2 Key Relationships

- One Card → Many Legs
- One Leg → One Vehicle + One Driver (time-bounded)
- One Vehicle/Driver → Many Legs (non-overlapping or admin-overridden)

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5. Business Rules

- Vehicles and Drivers are schedulable resources with calendars.
- Capacity filtering applies before vehicle selection.
- Vehicles are expected to logically start/end at institute (exceptions allowed with state tracking).
- Admin override is allowed anywhere but must be logged.
- Faculty cannot directly assign vehicles/drivers.
- Approved cards become read-only to Faculty except for clarification loops (if enabled).

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6. User Interface Requirements

6.1 Web (Primary for Transport Dept/Admin)

- Dashboard (Incoming, Today, Tomorrow, Future)
- Vehicle–Driver Roster with calendars
- Card Detail & Assignment Screens
- Conflict/Timeline view
- Audit/Reports

6.2 Mobile

- Faculty: card details, status, notifications
- Driver: duty calendar, assignment details, leave submission
- Admin (optional): quick approvals/changes

6.3 Visualization

- Timeline/Gantt-like vehicle view
- Map/route visualization (optional Phase-2)
- Icon-based leg visualization (mandatory)

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7. Non-Functional Requirements

7.1 Performance

- Real-time or near real-time updates (<5s propagation)
- Concurrent admin operations supported

7.2 Availability & Reliability

- ≥99.5% uptime target
- Transactional integrity for assignments and overrides

7.3 Security

- Role-based access control
- Secure APIs
- Audit-grade logging

7.4 Usability

- Minimal clicks for assignment
- Visual clarity for operational decisions
- Mobile-first for Faculty/Driver

7.5 Scalability

- Support increasing vehicles/drivers/cards without redesign

7.6 Maintainability

- Modular services (Cards, Scheduling, Assignment, Notification, Audit)
- Configurable rules where possible

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8. Integrations

- External Task Management System (Card assignment intake)
- Identity/SSO (optional)
- Notification services (Push/Email/SMS)

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9. Out of Scope (Phase-1)

- Fuel/billing management
- GPS/telematics live tracking
- Vendor taxi marketplace integrations
- Accounting/finance modules

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10. Future Enhancements (Indicative)

- Route optimization suggestions
- GPS live tracking & ETA
- Fuel/maintenance logs
- Analytics & utilization reports
- AI-assisted scheduling (advisory only)

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11. Acceptance Criteria (High Level)

- End-to-end card lifecycle works from intake → assignment → completion
- Multi-leg, multi-vehicle, multi-day missions executable
- Vehicle–driver daily planning and leave handling functional
- Visual journey UI implemented
- Notifications and audit logs verified
- Admin override available with logging

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12. Glossary

- **Card**: Transport request object
- **Mission**: Entire transport job for a request
- **Leg/Journey**: Segment within a mission
- **Date Bucket**: Grouping of missions by date
- **Override**: Admin action bypassing validations with audit

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📘 DETAILED SOFTWARE REQUIREMENTS SPECIFICATION

Transport Management System (TMS)

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1. System Overview

1.1 Objective

The Transport Management System (TMS) is designed to manage institutional transport operations involving:

• Faculty transportation
• Guest pickup/drop
• Multi-day transport missions
• Multi-vehicle scheduling
• Driver allocation
• Dynamic route planning
• Emergency override handling

The system must support **human decision-driven planning** rather than fully automated booking.

1.2 Architectural Philosophy

The system is built on three fundamental concepts:

A. Card Driven Workflow

All transport operations originate from a **Transport Card**

B. Mission Based Transport Model

One card = One mission
Mission may contain:
• Multiple journeys
• Multiple stops
• Multiple vehicles
• Multiple drivers

C. Resource Calendar Model

Vehicles and Drivers are treated as schedulable users with calendars.

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2. User Roles & Permissions

2.1 Faculty User

Permissions

• View assigned transport cards
• Fill transport details
• Track transport status
• Receive notifications

Restrictions

• Cannot assign vehicle
• Cannot modify approved transport

2.2 Transport In-Charge

Permissions

• Approve requests
• Schedule missions
• Assign vehicles
• Assign drivers
• Modify schedules

• Override system rules
• Close daily operations

2.3 Driver User

Permissions

• View assigned duties
• View route details
• Apply leave
• Receive notifications

2.4 Vehicle (System Resource)

Vehicle is NOT human user but behaves like:

• Has calendar
• Has availability
• Has assignment history
• Has capacity attributes

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3. External System Integration

3.1 Task Management System Integration

Transport system receives cards from external system when:

1. Faculty raises request
2. Internal authority approves
3. Faculty assigns card to Transport Department

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4. Transport Card Lifecycle

4.1 Card Status Flow

External Approved → Assigned to Transport
→ Details Pending
→ Submitted
→ Transport Approved
→ Date Bucketed
→ Vehicle Assigned
→ Driver Assigned
→ Active
→ Completed

4.2 Card Attributes

Mandatory Fields

- Request ID
- Faculty ID
- Travel date(s)
- Passenger count
- Journey type
- Pickup location
- Drop location

Optional Fields

- Guest details
- Return requirements
- Multi-day flag
- Special notes

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5. Faculty Module Detailed Requirements

5.1 Faculty Dashboard

Displays

• List of submitted cards
• Status indicator
• Date filter
• Search option

5.2 Transport Detail Form

Fields

Travel Information

• Travel Start Date
• Travel End Date
• One way / Two way / Multi day

Passenger Information

• Passenger count
• Guest names
• Contact details

Location Information

• Pickup location(s)
• Drop location(s)
• Multi stop toggle

Additional Inputs

• Special requirements
• Luggage details
• Accessibility requirement

5.3 Faculty Validation Rules

• Passenger count must be numeric
• Travel date cannot be past date
• Pickup and drop must be provided
• Multi-day travel requires return date

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6. Transport Department Dashboard

6.1 Incoming Request Dashboard

Displays

• New cards
• Emergency cards
• Pending clarification cards

Admin Actions

• Approve
• Reject
• Send clarification

6.2 Date Bucket Logic

System automatically categorizes cards:

IF travel date = today
→ Move to TODAY bucket

IF travel date = tomorrow
→ Move to TOMORROW bucket

ELSE
→ Move to FUTURE bucket

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7. Daily Planning Module

7.1 Vehicle Driver Mapping Screen

Displays

For each vehicle:

• Vehicle number
• Vehicle type
• Capacity
• Assigned driver
• Availability status

7.2 Driver Leave Management

Workflow

Driver submits leave
→ Admin approves leave
→ Vehicle driver slot becomes EMPTY

Admin must:
• Assign replacement driver
OR
• Mark vehicle idle

7.3 Day Closure Rules

Admin cannot close day until:

• All vehicle driver slots resolved

Admin Override Allowed:
• Allow vehicle idle
• Log override reason

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8. Mission Scheduling & Assignment Module

8.1 Mission Assignment Flow

Admin opens card
→ Click Assign Vehicle
→ Vehicle selection page opens

8.2 Vehicle Selection Screen

Displays

Free Vehicles

Vehicles with no conflicting timeline

Assigned Vehicles

Vehicles already scheduled with timeline visualization

8.3 Capacity Filter Rule

System auto filters vehicles:

Passenger Count → Eligible Vehicle Type

8.4 Timeline Display

For each vehicle:

• Start time from institute
• Destination arrival time
• Return start time
• Return arrival time

8.5 Reusing Vehicle Logic

Admin may assign vehicle already assigned IF:

• Timing overlap acceptable
• Route merge possible

System will:
• Allow manual time adjustment
• Send notification to affected users

8.6 Multi Vehicle Assignment

Each journey leg can have separate vehicle.

Example:
Leg 1 → Vehicle A
Leg 2 → Vehicle B

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9. Journey Visualization Module

System must visually display:

• Institute → Destination path
• Destination → Institute return path
• Stop markers
• Vehicle icon
• Timeline labels

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10. Supported Operational Scenarios

System MUST support:

• Multi pickup missions
• Multi drop missions
• Airport pickup missions
• One way drop missions
• Multi day stay missions
• Vehicle change mid mission
• Driver change mid mission
• Emergency reassignment
• Vehicle breakdown recovery

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11. Notification Module

11.1 Faculty Notifications

• Approval
• Vehicle assigned
• Timing changed
• Driver assigned

11.2 Driver Notifications

• Duty assigned
• Route details
• Duty modification

11.3 Admin Alerts

• Driver leave
• Conflict detection
• Emergency request

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12. Audit Module

System must record:

• All assignment history
• All schedule edits
• Override actions
• Approval timestamps
• User action logs

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13. Data Model (Developer Level)

13.1 Entities

User Table

UserID
Role
Contact details

Vehicle Table

VehicleID
VehicleType
Capacity
Status

Driver Table

DriverID
License
Availability

Card Table

CardID
FacultyID
TravelDate
PassengerCount
Status

Journey Table

JourneyID
CardID
StartLocation
EndLocation
StartTime
EndTime

Assignment Table

AssignmentID
JourneyID
VehicleID
DriverID

Leave Table

LeaveID
DriverID
LeaveDate

Audit Table

AuditID
ActionType
UserID
Timestamp

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14. Validation & Conflict Rules

System must detect:

• Vehicle double booking
• Driver double booking
• Capacity mismatch
• Driver leave conflicts

System must NOT block admin override.

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15. Non Functional Requirements

Performance

• Real time update required

Availability

• 99.5% uptime

Security

• Role based access

Scalability

• Must support expansion

16. Future Enhancements (Phase 2)

- GPS tracking
- Route optimization AI
- Fuel management
- Maintenance logs
- Analytics dashboard